

# The 2026–2027 Season Plan

*RANGER class · MATE ROV Competition*

Twelve months. Mid-June 2026 to Dauphin Island.

*Revised edition — competition-grade build, full season budget, and the administration track. Prepared May 2026.*

## Strategy

A new RANGER team will not out-build experienced teams in ten months. But it can out-prepare them on documentation, presentation, and safety — together, forty percent of the total score. So this plan front-loads skills, fundamentals, and the company's identity over the summer and fall, and treats the spring as execution against a known manual. Everything is in service of one deliverable, late April 2027: Dauphin Island.

This revision changes one thing about the build. The team is committing to a competition-grade, reliable platform — brushless thrusters, a professionally terminated tether, a sealed enclosure, an aluminum frame — chosen less for raw performance than for reliability and reuse. A vehicle that does not leak and does not fail mid-run protects the demonstration score, and every major component carries into future seasons. The budget rises accordingly, and is now honest: roughly \$7,600 to build, plus travel.

## The plan

### **Phase one — Foundation. Mid-June through July 2026.**

Six weeks. The goal is not a competition ROV. It is a team that exists, a working starter ROV in the water, and kids who have soldered something. Form the company — CEO, CFO, mechanical lead, electrical lead, software lead, communications lead. Build the SeaMATE Triggerfish kit as the team's first ROV, powered by a small 12 V sealed lead-acid battery the team supplies — the kit ships without a power source. Get every kid in a pool by week three. Lock pool access for the year, including one deep-water host. Hold a soldering bootcamp, an electronics workshop, and a project-notebook session, and set the safety culture on day one — every meeting opens with a safety brief. Add a waterproofing exercise: each kid designs a small sealed PVC capsule with a single cable penetration, drops it, recovers it, checks the paper towel inside, and iterates until it stays dry across three consecutive dives — the foundation skill for the float build in Phase 2. Start the community-outreach log on day one, and start the administrative work — registration, waivers, insurance, fundraising — in parallel.

|                                                                |                |
|----------------------------------------------------------------|----------------|
| SeaMATE Triggerfish ROV kit                                    | \$950          |
| 12 V SLA battery, charger, enclosure, and Powerpole leads      | \$125          |
| Two soldering stations, irons, and consumables                 | \$170          |
| Two multimeters and a hand-tool set                            | \$130          |
| Waterproofing capsule exercise (PVC, end caps, O-rings, epoxy) | \$50           |
| Heat-shrink, wire, zip ties, and build consumables             | \$55           |
| Ballast, trim weights, and flotation                           | \$20           |
| Bench-test bin and transport totes                             | \$40           |
| Safety gear (first aid kit, fire extinguisher, fume fan, mats) | \$100          |
| <b>Subtotal</b>                                                | <b>\$1,640</b> |

### **Phase two — Capability. August through mid-October 2026.**

Ten weeks. School starts. Settle into a weekly cadence: one weeknight, one weekend block. Each sub-team takes on a mini-project that maps to a real RANGER skill. Mechanical designs and 3D-prints a working gripper. Electrical builds a fused topside control box with proper terminations. Software writes a Python program that detects and counts objects in a webcam feed. The float team builds a v1 vertical profiler with a syringe buoyancy engine that hits 2.5 m and transmits depth — built on a NiMH battery, the chemistry the rules require, since alkaline and lithium-polymer are not permitted. Every member completes the Onshape learning path and captures a household object with photogrammetry. The prototypes are throwaway. The learning is the deliverable.

|                                                                                        |              |
|----------------------------------------------------------------------------------------|--------------|
| Float v1 components (microcontroller, MS5837 sensor, LoRa pair, syringe, NiMH battery) | \$260        |
| 3D-printer filament (community maker-space access assumed)                             | \$50         |
| Gripper and payload prototype materials                                                | \$60         |
| Topside control prototype (Raspberry Pi 4, gamepad, breadboard)                        | \$120        |
| USB camera for computer-vision work                                                    | \$30         |
| Outreach materials and workshop consumables                                            | \$125        |
| <b>Subtotal</b>                                                                        | <b>\$645</b> |

### Phase three — Design. Mid-October through mid-January 2027.

Twelve weeks. The 2027 manual drops around November. Within forty-eight hours of release, the team reads it together. Then a two-week decomposition: every task broken into requirements, scoring, and constraints. Trade studies follow — frame architecture, payload tools, camera choices, float comms protocol. Every decision earns a written rationale in the project notebook; those rationales become the technical documentation in March. CAD locks by mid-December. Long-lead parts ship before the holiday break — and this is the order that matters most, because Blue Robotics lead times stretch from February onward as every team orders at once.

This is where the competition vehicle is bought, and the plan commits to the winning-team build. Six Blue Robotics T200 thrusters — brushless, reliable, and reusable across multiple seasons. A pre-terminated tether, because faulty hand-made terminations are a top failure mode for new teams. A sealed Blue Robotics enclosure, an aluminum T-slot frame, and a regulated topside power supply. Every one of these is chosen for reliability and for reuse: the frame, enclosure, penetrators, tether, power supply, and thrusters all carry into the 2027-2028 vehicle.

|                                                          |                |
|----------------------------------------------------------|----------------|
| Six Blue Robotics T200 thrusters                         | \$1,506        |
| Six Basic ESC R3 motor controllers                       | \$240          |
| Blue Robotics 4-inch watertight enclosure assembly       | \$250          |
| Aluminum T-slot frame extrusion and hardware             | \$250          |
| Pre-terminated competition tether (Fathom-class)         | \$350          |
| Topside power supply (regulated, 48 V)                   | \$300          |
| Raspberry Pi 5 (ROV) and Arduino Mega (sensor interface) | \$190          |
| Two underwater cameras (USB, DIY housings)               | \$100          |
| WetLink penetrators, eight                               | \$105          |
| Main fuse and ANL/MIDI fuse holder                       | \$20           |
| Clamp ammeter (full-load-amp measurement)                | \$70           |
| Float v2 refinement parts                                | \$80           |
| Contingency buffer                                       | \$100          |
| <b>Subtotal</b>                                          | <b>\$3,561</b> |

*Note on the thruster decision. Six T200 thrusters are the largest single line in the plan. The order is placed in December, but it is worth a deliberate go/no-go in the fall: confirmed if the funding is in hand, deferred to bilge-pump conversions (about a fifth of the cost) if it is not. Either path uses the same frame, geometry, and control architecture — only the motors and drivers differ.*

#### **Phase four — Build. Mid-January through February 2027.**

Six weeks. Construction. Tether, topside control box, ROV frame, electronics integration, float build, and the pneumatic manipulator. Build a topside bench replica that mirrors the ROV electronics — it lets software develop in parallel with hardware and gives the team something to demonstrate while the real vehicle stays safe. Plan for two weeks of “it doesn’t work” before it works. Bare frame wet by end of January. Electronics integrated by mid-February. First full-system test by end of February. The fluid-power quiz and safety pre-inspection tracking begin now.

|                                                               |              |
|---------------------------------------------------------------|--------------|
| Pneumatic manipulator (cylinder, solenoid, regulator, tubing) | \$175        |
| Topside bench replica (parallel software development)         | \$70         |
| Spare ESCs and electronics backups                            | \$80         |
| Rework consumables (epoxy, sealants, O-rings)                 | \$80         |
| Mounting hardware, fasteners, brackets                        | \$80         |
| Replacement-parts buffer (one major break)                    | \$150        |
| Display-board base materials                                  | \$60         |
| Misc build consumables                                        | \$50         |
| <b>Subtotal</b>                                               | <b>\$745</b> |

#### **Phase five — Test and document. March 2027.**

Four weeks. Full mission rehearsals at the deep-water pool every weekend. Time each task. Identify what breaks; fix it. Lock the SIDs — electrical, fluid power, non-ROV device. Write the JSEA, focused on poolside operations rather than shop safety. Draft the technical documentation — student-written, with any AI assistance disclosed, as the rules require. Begin weekly mock engineering presentations with an outside adult acting as a judge; the presentation is live and oral, with no slide deck permitted. Prepare the marketing display, including the digital copy now required.

|                                                   |              |
|---------------------------------------------------|--------------|
| Critical spares (ESC, penetrator, sensor backups) | \$200        |
| Marketing display printing (36 × 48) and mounting | \$130        |
| Documentation printing and binding                | \$50         |
| Backup hardware (spare Raspberry Pi, sensors)     | \$100        |
| <b>Subtotal</b>                                   | <b>\$480</b> |

## Phase six — Polish and compete. April 2027 through the regional.

Documentation submission deadlines hit in early April. Confirm DISL’s exact dates. Print the marketing display, thirty-six by forty-eight inches. Five engineering-presentation rehearsals, minimum, on top of the weekly mock Q&A that began in January. Pack the trailer. Drive to Dauphin Island, late April 2027. Note that regional registration opens in mid-February — the team must be registered well before it travels.

|                                              |              |
|----------------------------------------------|--------------|
| Regional registration fee                    | \$250        |
| Final marketing-display reprint and handouts | \$60         |
| Team shirts (eight at \$20)                  | \$160        |
| Day-of miscellaneous                         | \$50         |
| <b>Subtotal</b>                              | <b>\$520</b> |

## The parallel track — administration, registration, and funding

Three workstreams run alongside the build from day one. They belong to the parents, not the students, and a missed deadline on any of them can end the season as surely as a leak.

**Registration and compliance.** Register the company with MATE. The Northern Gulf Coast regional registration window opens in mid-February 2027 — the team must be registered before it closes. Collect liability waivers, photo-and-media releases, and a medical and emergency form for every student. Confirm the program carries liability insurance; the pool host will likely ask for proof. Check whether the team’s affiliation requires youth-protection background checks for adult mentors.

**The DISL relationship.** Contact Rachel McDonald at the Dauphin Island Sea Lab early — to confirm the 2027 competition dates, the pool’s depth and dimensions, and exactly which documents the regional requires.

**Funding.** The build costs roughly \$7,600, and travel to the regional adds roughly \$2,000 more. That money has to be raised. Start a sponsor list and a fundraising plan in the summer, not the spring.

## Skills the team will leave with

By April, every kid will have hands-on with: soldering and basic electronics, fusing and full-load-amp measurement, waterproofing — O-rings, epoxy potting, penetrators — 3D CAD in Onshape, 3D printing, Python with either Arduino or Raspberry Pi, OpenCV for image recognition, photogrammetry, pressure and depth sensors, wireless RF, pneumatics, safety culture and hazard analysis, basic project management, technical writing, and public speaking with live Q&A. Specialists will go deeper in their lanes.

## Budget summary

*Working totals based on 2026 vendor pricing for Blue Robotics thrusters and enclosures, a Fathom-class tether, and the SeaMATE kit. Excludes pool access (donated facility) and project notebooks (kids bring their own). Travel is now estimated rather than excluded.*

|                                               |                 |
|-----------------------------------------------|-----------------|
| Phase one — Foundation                        | \$1,640         |
| Phase two — Capability                        | \$645           |
| Phase three — Design                          | \$3,561         |
| Phase four — Build                            | \$745           |
| Phase five — Test and document                | \$480           |
| Phase six — Polish and compete                | \$520           |
| <b>Build total</b>                            | <b>\$7,591</b>  |
| Travel and lodging to the regional (estimate) | ~\$2,000        |
| <b>All-in season cost</b>                     | <b>~\$9,600</b> |

*The travel estimate assumes carpooling and shared lodging for the six-hour drive to Dauphin Island; a fuller trip runs higher. If the team wins the regional and advances to the World Championship, budget a further \$3,000–\$5,000 — upside, not baseline.*

## Milestones

|                             |                                                                                                                        |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>End of July 2026</b>     | Starter ROV in the water. Deep-pool access confirmed. Company formed. Registration paperwork and fundraising underway. |
| <b>November 2026</b>        | 2027 manual released. Full-team read-through within forty-eight hours.                                                 |
| <b>Mid-December 2026</b>    | CAD locked. Long-lead parts — thrusters, enclosure, tether — ordered before the holiday break.                         |
| <b>Mid-February 2027</b>    | Regional registration opens. The team registers.                                                                       |
| <b>End of February 2027</b> | First full-system pool test.                                                                                           |
| <b>Early April 2027</b>     | Technical-documentation submission deadline. Confirm with DISL.                                                        |
| <b>Late April 2027</b>      | Northern Gulf Coast Regional. Dauphin Island Sea Lab.                                                                  |

## Risks, and how to defuse them

Two failure modes kill new RANGER teams. The first is not getting in the water enough. Fix this by locking pool access in June, not October. The second is under-investing in the engineering-and-communications track because the ROV isn't done yet. Fix this by starting the marketing display and the technical-documentation skeleton in August, even when there is little to put in them.

A third risk lives at the parent level: over-functioning. The rule for the year is — teach a skill once, then step back and let them fail. The kids learning to recover from failure is what makes the engineering-presentation Q&A go well in April.

Two risks are new to this revision. **Administrative deadlines:** a missed registration date or an unsigned waiver can end the season as surely as a leak. Treat the parallel track as a real workstream with a named owner, not an afterthought. **Supply lead times:** Blue Robotics lead times stretch from February as every team orders at once. Place the long-lead order before the holiday break — which is why CAD locks in mid-December.

## Where to start

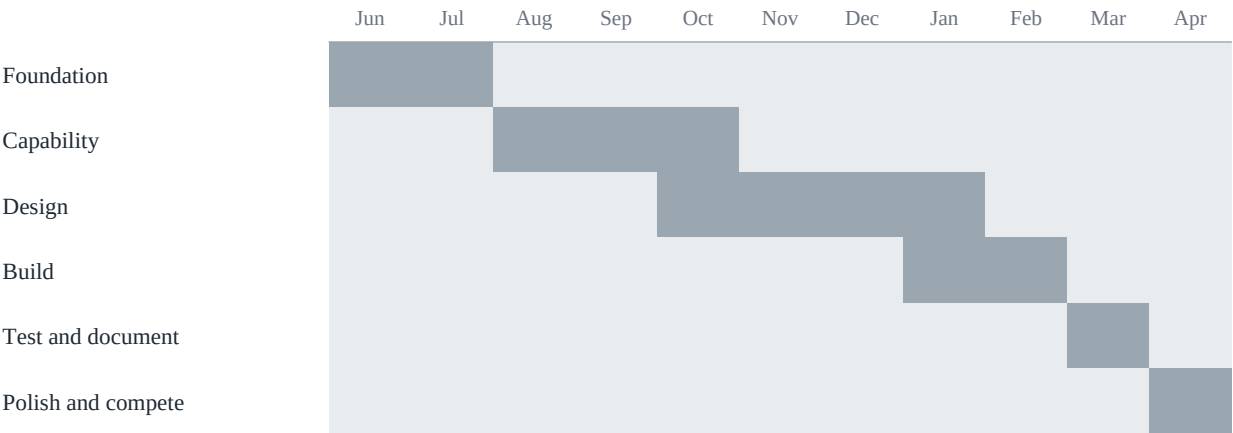
If we do nothing else in June: buy the SeaMATE Triggerfish kit, get six kids soldering, and put it in the water within three weeks. Start the outreach log, the waivers, and the sponsor conversations the same month. Everything else in this plan is downstream of that.

## A note on the second season

The 2027 vehicle is built to be inherited. The aluminum frame, the sealed enclosure, the penetrators, the tether, the topside power supply, and the T200 thrusters all carry into the 2027-2028 build — the season's largest costs are one-time. A second season reuses the platform and iterates one subsystem, the pattern every multi-year winning program follows. Realistically, Season 2 runs closer to \$3,500–\$4,700 all-in against this season's roughly \$9,600 — the difference is simply not building the platform twice. The most important thing to protect for Season 2 is not hardware but people: cross-train two students per role, because graduation, not cost, is what ends young programs.

# Appendix — Integrated Master Schedule

One page. Six phases. Eleven months. The bars show duration; the milestones below mark the dates above. Pin it to a wall.



## Milestones

|          |                                     |
|----------|-------------------------------------|
| End Jul  | Starter ROV wet                     |
| Nov      | 2027 manual released                |
| Mid-Dec  | CAD locked, long-lead parts ordered |
| Mid-Feb  | Regional registration opens         |
| End Feb  | First full-system test              |
| Late Apr | Dauphin Island regional             |